

PROVATINE CAPSULE 20MG

DESCRIPTION:

A green/ cream, size '3' capsule with markings 'DUO 861'.

COMPOSITION:

Each capsule contains Fluoxetine HCl equivalent to Fluoxetine 20mg.
Gelatin for Capsule: Halal certified bovine source.

PHARMACODYNAMICS:

The antidepressant and anti-obsessional action of fluoxetine is presumed to be linked to its inhibition of CNS neuronal uptake of serotonin. Studies at clinically relevant doses in man have demonstrated that fluoxetine blocks the uptake of serotonin, but not of noradrenaline, into human platelets. Studies in animals also suggest that fluoxetine is a much more potent uptake inhibitor of serotonin than of noradrenaline. Antagonism of muscarinic, histaminergic, and α 1-adrenergic receptors has been hypothesised to be associated with various anticholinergic, sedative, and cardiovascular effects of classical tricyclic antidepressant drugs. Fluoxetine binds to these and other membrane receptors from brain tissue much less potently in vitro than do the tricyclic drugs.

PHARMACOKINETICS:

Absorption: In man, following a single oral 40 mg dose, peak plasma concentrations of fluoxetine from 15 to 55 ng/mL are observed after 6 to 8 hours. Fluoxetine is 80 to 95% absorbed following oral administration. There is a linear dose proportionality for the absorption of fluoxetine over the therapeutic dose range. Food does not appear to affect the systemic bioavailability of fluoxetine, although it may delay its absorption inconsequentially. Thus, fluoxetine may be administered with or without food.

Distribution: The volume of distribution for fluoxetine is estimated at 30-40 L/kg.

Protein Binding: Over the concentration range from 200 to 1,000 ng/mL, approximately 94.5% of fluoxetine is bound in vitro to human serum proteins, including albumin and α -1-glycoprotein. The interaction between fluoxetine and other highly protein-bound drugs has not been fully evaluated, but may be important (see Precautions).

Metabolism: Fluoxetine is extensively metabolised in the liver to norfluoxetine and a number of other, unidentified metabolites. The only identified active metabolite, norfluoxetine, is formed by demethylation of fluoxetine. In animal models, norfluoxetine's potency and selectivity as a serotonin uptake blocker are essentially equivalent to fluoxetine's.

Multiple cytochrome P450 isoenzymes, including CYP2D6, are responsible for the conversion of fluoxetine to norfluoxetine; thus other nonsaturable oxidative pathways (i.e. non-2D6 pathways) contribute considerably to norfluoxetine formation (see Interactions with Other Drugs).

Excretion: The primary route of elimination appears to be hepatic metabolism to inactive metabolites excreted by the kidney.

Clinical Issues Related to Metabolism/Excretion: The complexity of the metabolism of fluoxetine has several consequences which may potentially affect fluoxetine's clinical use.

Accumulation and Slow Elimination: The relatively slow elimination of fluoxetine (elimination half-life of 1 to 3 days after acute administration and 4 to 6 days after chronic administration) and its active metabolite, norfluoxetine (elimination half-life of 4 to 16 days after acute and chronic administration), leads to significant accumulation of these active species in chronic use. After 30 days of dosing at 40 mg/day, plasma concentrations of fluoxetine in the range of 91 to 302 ng/mL and norfluoxetine in the range of 72 to 258 ng/mL have been observed. Plasma concentrations of fluoxetine were higher than those predicted by single dose studies, presumably because fluoxetine's metabolism is not proportional to dose. Norfluoxetine, however, appears to have linear pharmacokinetics. Its mean terminal half-life after a single dose was 8.6 days and after multiple dosing was 9.3 days. Thus, even if patients are given a fixed dose, steady state plasma concentrations are only achieved after continuous dosing for weeks. Nevertheless, plasma concentrations do not appear to increase without limit. Specifically, patients receiving fluoxetine at doses of 40 to 80 mg/day over periods as long as 3 years exhibited, on average, plasma concentrations similar to those seen among patients treated for 4 or 5 weeks.

INDICATION:

Provatine is indicated for the treatment of:

Major depression.

Obsessive Compulsive Disorder.

Premenstrual Dysphoric Disorder (PMDD) as defined by DSM-IV criteria.

The essential features of PMDD, according to the Diagnostic and Statistical Manual-4th edition (DSM-IV) include markedly depressed mood, anxiety or tension, affective lability and persistent anger or irritability. Other features include decreased interest in usual activities, difficulty concentrating, lack of energy, change in appetite or sleep and feeling out of control. Physical symptoms associated with PMDD include breast tenderness, headache, joint and muscle pain, bloating and weight gain. These symptoms occur regularly during the luteal phase and remit within a few days following onset of menses; the disturbance markedly interferes with work or school or with usual social activities and relationships with others.

RECOMMENDED DOSAGE:

Depression:

The usual recommended initial dose is 20 mg/day, administered in the morning.

If no clinical improvement is observed after several weeks, a dose increase may be considered. Doses above 20 mg/day should be administered twice daily (morning and noon) and should not exceed a maximum dose of 80 mg/day. As with other antidepressants, the full antidepressant effect may be delayed until four weeks of treatment or longer. As with many other medications, a lower or less frequent dosage should be used in patients with renal and/or hepatic impairment. A lower or less frequent dosage should also be considered for elderly patients and patients with concurrent disease or on multiple medications.

Obsessive Compulsive disorder:

Initial treatment: 20 mg/day administered in the morning is recommended as the initial dose. If insufficient clinical improvement is observed after several weeks, a dose increase may be considered. The full therapeutic effect may be delayed until five weeks of treatment or longer.

Doses above 20 mg/day may be administered on a once daily (morning) or twice daily (morning and afternoon) schedule. A dose range of 20 to 60 mg/day is recommended however doses of up to 80 mg/day have been well tolerated in open studies of obsessive-compulsive disorder (OCD). The maximum fluoxetine dose should not exceed 80 mg/day. A lower or less frequent dose should be used in patients with renal and/or hepatic impairment. A lower or less frequent dosage should also be considered for elderly patients with concurrent disease or on multiple medications.

Maintenance/continuation treatment: While there are no systematic studies that answer the question of how long to continue fluoxetine therapy, OCD is a chronic condition and is reasonable to consider continuation for responding patient. However, dosage adjustments should be made to maintain the patient on the lowest effective dosage and patients should be periodically reassessed to determine the need for treatment.

Impaired hepatic function: as might be predicted from its primary site of metabolism, hepatic impairment can affect the elimination of fluoxetine. The elimination half-life of fluoxetine was prolonged in a study of patients with cirrhosis, with a mean of 7.6 days compared to the range of two to three days seen in subjects without hepatic disease; norfluoxetine elimination was also delayed, with a mean duration of 12 days for patients with cirrhosis compared to the range of seven to nine days in normal subjects. This suggests that the use of fluoxetine in patients with hepatic disease must be approached with caution. If fluoxetine is administered to patients with hepatic disease, a lower or less frequent dose should be used.

Impaired renal function: In a single dose studies the pharmacokinetics of fluoxetine and norfluoxetine were similar among subjects with all levels of impaired renal function, including anephric patients on chronic hemodialysis. However, with chronic administration, additional accumulation of fluoxetine or its metabolites (possibly including some not yet identified) may occur in patients with severely impaired renal function, and use of a lower or less frequent dose is advised.

Premenstrual Dysphoric Disorder (PMDD): The recommended dose of fluoxetine for the treatment of PMDD is 20 mg/day given continuously (every day of the menstrual cycle) or intermittently (defined as starting a daily dose 14 days prior to the anticipated onset of menstruation through the first full day of menses and repeating with each new cycle).

ROUTE OF ADMINISTRATION:

Oral

CONTRAINDICATIONS:

Known hypersensitivity to the drug.

Monoamine oxidase inhibitors: There have been reports of serious, sometimes fatal reactions (including hyperthermia, rigidity, myoclonus, autonomic instability with possible rapid fluctuations of vital signs, and mental status changes that include extreme agitation progressing to delirium and coma) in patients receiving fluoxetine in combination with a monoamine oxidase inhibitor (MAOI), and in patients who have recently discontinued fluoxetine and are then started on an MAOI. Some cases presented with features resembling neuroleptic malignant syndrome.

Therefore, fluoxetine hydrochloride should not be used in combination with a MAOI (selective, reversible or irreversible) or within a minimum of 14 days of discontinuing therapy with an MAOI. Since fluoxetine and its major metabolite have very long elimination half-lives, at least five weeks (perhaps longer, especially if fluoxetine has been prescribed chronically and/or at higher doses (see Pharmacokinetics – Accumulation and Slow Elimination) should elapse after discontinuing Provatine treatment before starting a MAOI. Limited reports suggest that orally administered cyproheptadine or intravenously administered dantrolene may benefit patients experiencing such reactions. Animal studies also suggest that cyproheptadine may be beneficial.

WARNINGS & PRECAUTIONS:**Suicidality in Children and Adolescents**

- Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.
- Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need.
- Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.
- Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.
- Provatine Capsule is not approved for use in pediatric patients.

Fluoxetine and other selective serotonin reuptake inhibitors (SSRIs) that undergo hepatic metabolism and renal excretion should be used with caution and in reduced doses in patients with impaired hepatic or renal function and some authorities recommend that they should not be used in severe hepatic or renal failure. Because of this epileptogenic effects SSRIs should be used with caution in patients with epilepsy or a history of such disorders (and should be avoided if the epilepsy is poorly controlled). Fluoxetine may alter glycaemic control and therefore caution is also warranted in diabetic subjects. Fluoxetine should be discontinued in patients who develop a rash since systemic effects, possibly related to vasculitis, have occurred in such patients. Patients should be closely monitored during early therapy until improvement in depression is observed because suicide is an inherent risk in depressed patients. If SSRIs are given for the depressive component of bipolar disorder, mania may be precipitated.

SSRIs may impair performance of skilled tasks and, if affected, patients should not drive or operate machinery. Some authorities recommend reduced or less frequent dosage of SSRIs for elderly patients. Significant amounts of fluoxetine are distributed into breast milk and the manufacturers have reported that adverse effects have been observed in a breast fed infant; consequently, fluoxetine is not recommended that they should be avoided during breast feeding (fluoxetine and paroxetine) or used with caution (citalopram). SSRIs should be withdrawn gradually to reduce the risk of withdrawal symptoms.

Altered Appetite and Weight: Significant weight loss, especially in underweight depressed patients, may be an undesirable result of treatment with fluoxetine hydrochloride. In controlled clinical trials, approximately 9% of patients treated with fluoxetine hydrochloride experienced anorexia. This incidence is approximately sixfold that seen in placebo controls. A weight loss of greater than 5% of body weight occurred in 13% of fluoxetine hydrochloride treated patients compared to 4% of placebo and 3% of tricyclic antidepressant treated patients. However, only rarely have fluoxetine hydrochloride patients been discontinued for weight loss.

Provatine has not been evaluated or used to any appreciable extent in patients with a recent history of myocardial infarction or unstable heart disease. Patients with these diagnoses were systematically excluded from clinical studies during the product's premarket testing. However, the electrocardiograms of 312 patients who received fluoxetine hydrochloride in double-blind trials were retrospectively evaluated; no conduction abnormalities that resulted in heart block were observed. The mean heart rate was reduced by approximately 3 beats/min.

Platelet Function: There have been reports of altered platelet function and/or abnormal results from laboratory studies in patients taking fluoxetine. While there have been reports of abnormal bleeding in several patients taking fluoxetine, it is unclear whether fluoxetine had a causative role.

Sexual dysfunction: Selective serotonin reuptake inhibitors (SSRIs)/ serotonin norepinephrine reuptake inhibitors (SNRIs) may cause symptoms of sexual dysfunction. There have been reports of long-lasting sexual dysfunction where the symptoms have continued despite discontinuation of SSRIs/SNRIs.

INTERACTION WITH OTHER MEDICAMENTS:

As with all drugs, the potential for interaction by a variety of mechanisms (i.e. pharmacodynamic, pharmacokinetic drug inhibition or enhancement) is a possibility.

Drugs metabolised by cytochrome P450 2D6: Approximately 3 to 10% of the normal population has a genetic defect that leads reduced levels of activity of cytochrome P450 2D6 (CYP 2D6). Such individuals have been referred to as poor metabolisers of drugs such as dextromethorphan and tricyclic antidepressants. Many drugs, such as most antidepressants including fluoxetine and other selective uptake inhibitors of serotonin, are metabolised by this isoenzyme, thus both the pharmacokinetic properties and relative proportion of metabolites are altered in poor metabolisers. Fluoxetine, like other agents that are metabolised by CYP 2D6, inhibits the activity of this isoenzyme and thus may make normal metabolisers resemble poor metabolisers. Therapy with medications that are predominantly metabolised by CYP 2D6 and that have a relatively narrow therapeutic index (e.g. flecainide, tricyclic antidepressants) should be initiated at the low end of the dose range if a patient is taking fluoxetine. Concurrently or has taken it in the previous five weeks. A washout period of two weeks should be allowed when changing from medication with a tricyclic antidepressant to fluoxetine.

Highly protein bound drugs: Because fluoxetine is tightly bound to plasma protein, the administration of fluoxetine to a patient taking another drug that is tightly bound to protein (e.g. warfarin) may cause a shift in plasma concentrations potentially resulting in an adverse effect. Conversely, adverse effects may result from displacement of protein bound fluoxetine by other tightly bound drugs.

Central nervous system active drugs: The risk of using fluoxetine in combination with other CNS active drugs has not been systematically evaluated. Data have been derived from circumstances, which do not directly reflect the clinical setting. The clinical significance of in vitro and individual case report data is unknown. Nonetheless, caution is advised if the concomitant administration of fluoxetine and such drugs is required. In evaluating individual cases, consideration should be given to using lower initial doses of the concomitantly administered drugs, using conservative titration schedules and monitoring of clinical status.

Anticonvulsants: Patients on stable doses of phenytoin and carbamazepine have developed elevated plasma anticonvulsant concentrations and clinical anticonvulsant toxicity following initiation of concomitant fluoxetine treatment.

Antipsychotics: Some evidence suggests a possible pharmacodynamic and/or pharmacokinetic interaction between some SSRIs and some antipsychotics, including possible elevation of blood levels of haloperidol and clozapine. A single case report has suggested possible additive effects of pimozide and fluoxetine leading to bradycardia.

Benzodiazepines: The half-life of concurrently administered diazepam may be prolonged in some patients and coadministration of alprazolam may result in increased alprazolam concentrations.

Lithium: There have been reports of both increased and decreased lithium levels when lithium was used concomitantly with fluoxetine. Cases of lithium toxicity and increased serotonergic effects have been reported. Lithium levels should be monitored when these drugs are administered concomitantly.

Other antidepressants: In two studies, previously stable plasma levels of imipramine and desipramine have increased greater than two to ten times when fluoxetine has been administered in combination. This influence may persist for three weeks or longer after fluoxetine has been discontinued. Thus, the dose of tricyclic antidepressant may need to be reduced and plasma tricyclic antidepressant concentrations may need to be monitored temporarily when fluoxetine is coadministered or has been recently discontinued.

Hypericum perforatum (St John's wort): Fluoxetine has been identified as a drug which may have a clinically significant pharmacodynamic and pharmacokinetic interactions possibly involving the cytochrome P450 isoenzymes, with products containing *Hypericum perforatum* (St John's wort).

Instructions to patients: Doctors are advised to discuss the following issues with patients for whom they prescribe fluoxetine. Because fluoxetine hydrochloride may impair judgment, thinking or motor skills, patients should be advised to avoid driving a car or operating hazardous machinery until they are reasonably certain that their performance is not affected. Patients should be advised to inform their doctor if they are taking or plan to take any prescription or over-the-counter drugs, or alcohol. Patients should be advised to notify their doctor if they become pregnant or intend to become pregnant during therapy, if they are breast feeding an infant or if they develop a rash or hives.

Serotonergic drugs: Coadministration with serotonergic drugs (e.g. tramadol, sumatriptan) may lead to an increase of 5-HT-associated effects.

Electroconvulsive Therapy: There are no clinical studies establishing the benefit of the combined use of ECT and Provatine. There have been some reports of prolonged seizures in patients on Provatine receiving ECT treatment.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy:

This drug crosses the placenta and may produce effects in newborn infants.

Teratogenic effects: Reproduction studies have been performed in rats and rabbits at doses of up to 12.5 and 15 mg/kg/day and have revealed no evidence of harm to the fetus due to fluoxetine hydrochloride. There are, however, no adequate and well controlled studies in pregnant women.

Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed. **Labour and delivery:** The effect of fluoxetine on labour and delivery in humans is unknown. Drugs which, owing to their pharmacological effects, have caused or may be suspected of causing, harmful effects on the human foetus or neonate without causing malformations. These effects may be reversible. Accompanying text above should be consulted for further details.

Observational data indicate an increased risk (less than 2-fold) of postpartum haemorrhage following SSRI/SNRI exposure within the month prior to birth.

Use in Lactation:

Because fluoxetine is excreted in human milk, breast feeding while on fluoxetine is not recommended. In one breast milk sample, the concentration of fluoxetine plus norfluoxetine was 70.4 nanogram/mL. The concentration in the mother's plasma was 295.0 nanogram/mL. No adverse effects on the infant were reported. In another case, an infant breastfed by a mother on fluoxetine 20 mg developed crying, sleep disturbance, vomiting and watery stools. The infant's plasma drug levels were fluoxetine 340 nanogram/mL and norfluoxetine 208 nanogram/mL on the second day of feeding.

Use in children: Safety and effectiveness in children have not been established.

Effect on ability to drive or operate machinery: Patients should be cautioned about operating hazardous machinery or driving a car until they are reasonably certain that treatment with fluoxetine does not affect them adversely.

SIDE EFFECTS:

Selective serotonin reuptake inhibitors (SSRIs) such as fluoxetine are less sedative than tricyclic antidepressants and have few antimuscarinic or cardiotoxic effects. Adverse effects reported with fluoxetine include dry mouth and gastro-intestinal disturbances such as nausea, vomiting, dyspepsia, constipation, and diarrhea. Anorexia and weight loss may also occur. Neurological side-effects have included either anxiety, restlessness, nervousness, and insomnia or drowsiness and fatigue; headache, tremor, dizziness, convulsions, extrapyramidal effects, and that some SSRIs may be associated with increased suicidal ideation. Excessive sweating and pruritus or skin rashes such as urticaria have also been reported. In some patients who have developed rashes while taking fluoxetine, systemic events involving the lungs, kidneys, or liver, and possibly related to vasculitis, have developed; it has therefore been advised that fluoxetine therapy should be discontinued in any patient who develops a skin rash. Hyponatremia, possibly due to inappropriate secretion of antidiuretic hormone, has been associated with the use of antidepressants, particularly in the elderly. Abnormal liver function tests have been reported. In overdose,

nausea, vomiting, and excitation of the CNS are considered to be prominent features; death has been reported.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Treatment of overdosage with a selective serotonin reuptake inhibitor (SSRI) involves emesis induction or gastric lavage followed by symptomatic and supportive therapy. The use of activated charcoal by mouth as an adjunct to gastric lavage has been suggested. Dialysis, haemoperfusion, exchange transfusion, and measures to increase urine production are considered unlikely to be of benefit.

STORAGE CONDITIONS:

Store below 30°C. Protect from light.

Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Packs of 100's and 3 X 10's & 10 X 10's in blister pack.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang,
Selangor, MALAYSIA.